

Instrumental Variable Regression

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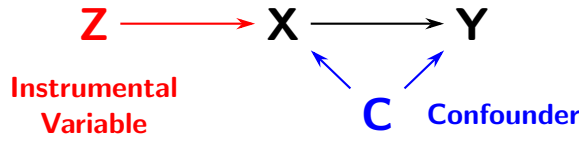
1 Problem Setting

We consider a simple regression problem

$$y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon, \quad (1)$$

where $\mathbf{x}, \boldsymbol{\beta} \in \mathbb{R}^{d_x}$, ε is a random variable with $\mathbb{E}(\varepsilon) = 0$ and $\text{Var}(\varepsilon) = \sigma^2$. In the simplest setting, we assume that the explanatory variable $\mathbf{x}$ is **exogenous**, i.e., $\mathbf{x}$ is uncorrelated with ε . However, in many real applications, this assumption may not hold. Instead, there may be a **confounder** $\mathbf{c} \in \mathbb{R}^{d_c}$ that has causal effects on both $\mathbf{x}$ and y , thereby violating the exogeneity assumption and causing **endogenous**.

Such a setting, where endogeneity is present, leads to the problem of Instrumental Variable (IV) regression. In this setting, the **instrumental variable** $\mathbf{z} \in \mathbb{R}^{d_z}$ is introduced to eliminate the endogeneity caused by $\mathbf{c}$. The instrumental variable $\mathbf{z}$ is uncorrelated with the confounder $\mathbf{c}$. The causal relationships of all variables can be represented using a directed acyclic graph (DAG):



We finally formulate the IV regression problem. The true model is

$$y = \boldsymbol{\theta}^\top \mathbf{x} + \underbrace{\boldsymbol{\alpha}^\top \mathbf{c} + \varepsilon_y}_{\varepsilon}, \quad (2)$$

$$\mathbf{x} = \boldsymbol{\gamma}^\top \mathbf{z} + \boldsymbol{\beta}^\top \mathbf{c} + \boldsymbol{\varepsilon}_x, \quad (3)$$

where $\boldsymbol{\theta} \in \mathbb{R}^{d_x}$, $\boldsymbol{\alpha} \in \mathbb{R}^{d_c}$, $\boldsymbol{\gamma} \in \mathbb{R}^{d_z \times d_x}$, $\boldsymbol{\beta} \in \mathbb{R}^{d_c \times d_x}$ are parameters, $\varepsilon_y \in \mathbb{R}$ and $\boldsymbol{\varepsilon}_x \in \mathbb{R}^{d_x}$ are random noise and are independent with all other variables. Our goal is to estimate $\boldsymbol{\theta}$ from the data $\{(\mathbf{z}_i, \mathbf{x}_i, y_i)\}_{i=1}^n$. We denote $X = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top \in \mathbb{R}^{n \times d_x}$, $Z = (\mathbf{z}_1, \dots, \mathbf{z}_n)^\top \in \mathbb{R}^{n \times d_z}$, $Y = (y_1, \dots, y_n)^\top \in \mathbb{R}^{d_y}$ as the data matrices, and $C = (\mathbf{c}_1, \dots, \mathbf{c}_n)^\top \in \mathbb{R}^{n \times d_c}$ as the unknown confounder matrix.

2 Solving the IV Regression Problem

2.1 Ordinary Least Squares

The most straightforward way to estimate $\boldsymbol{\theta}$ is via the Ordinary Least Squares (OLS) method. However, due to the endogeneity of $\mathbf{x}$, the OLS estimator no longer retains its desirable statistical properties, as it can be derived that

$$\hat{\boldsymbol{\theta}}_{\text{OLS}} = (X^\top X)^{-1} X^\top Y = \boldsymbol{\theta} + (X^\top X)^{-1} X^\top C \boldsymbol{\alpha} \xrightarrow{p} \boldsymbol{\theta} + \Sigma_{XX}^{-1} \Sigma_{XC} \boldsymbol{\alpha}. \quad (4)$$

Since $\mathbf{x}$ and $\mathbf{c}$ is correlated, we have $\Sigma_{XC} \neq 0$, and hence the OLS estimator $\hat{\boldsymbol{\theta}}_{\text{OLS}}$ is biased and inconsistent.

2.2 Two-Stage Least Squares

A better way to estimate $\boldsymbol{\theta}$ is via the Two-Stage Least Squares (2SLS). The main idea of 2SLS is to use the instrumental variable $\mathbf{z}$ to eliminate the endogeneity of $\mathbf{x}$. Specifically, we first use $\mathbf{z}$ to regress $\mathbf{x}$, and then use the estimated values of $\mathbf{x}$ to regress y . Since $\mathbf{z}$ is uncorrelated with $\mathbf{c}$, it can be used to isolate the exogenous variation in $\mathbf{x}$ and thereby remove the confounding effect in estimating the relationship between $\mathbf{x}$ and y .

The OLS estimate of $\mathbf{x}_i$ can be given by

$$\hat{\mathbf{x}}_i = \hat{\gamma}_{\text{OLS}}^\top \mathbf{z}_i = X^\top Z (Z^\top Z)^{-1} \mathbf{z}_i, \quad (5)$$

and the 2SLS estimator of $\boldsymbol{\theta}$ can be derived as

$$\hat{\boldsymbol{\theta}}_{\text{2SLS}} = (\hat{X}^\top \hat{X})^{-1} \hat{X}^\top Y = (X^\top P_Z X)^{-1} X^\top P_Z Y, \quad (6)$$

where $P_Z = Z(Z^\top Z)^{-1} Z^\top$ denotes the projection matrix onto the column space of Z , which extracts the component of X explained by the instrumental variable, and thereby isolates the exogenous variation in X . Similar to (4), we have

$$\hat{\boldsymbol{\theta}}_{\text{2SLS}} = \boldsymbol{\theta} + (X^\top P_Z X)^{-1} X^\top P_Z C \boldsymbol{\alpha}. \quad (7)$$

For the error term, we have

$$\begin{aligned} (X^\top P_Z X)^{-1} X^\top P_Z C \boldsymbol{\alpha} &= (X^\top Z (Z^\top Z)^{-1} Z^\top X)^{-1} X^\top Z (Z^\top Z)^{-1} Z^\top C \boldsymbol{\alpha} \\ &= \left(\left(\frac{X^\top Z}{n} \right) \left(\frac{Z^\top Z}{n} \right)^{-1} \left(\frac{Z^\top X}{n} \right) \right)^{-1} \left(\frac{X^\top Z}{n} \right) \left(\frac{Z^\top Z}{n} \right)^{-1} \left(\frac{Z^\top C}{n} \right) \boldsymbol{\alpha} \\ &\xrightarrow{p} (\Sigma_{XZ} \Sigma_{ZZ}^{-1} \Sigma_{ZX})^{-1} \Sigma_{XZ} \Sigma_{ZZ}^{-1} \underbrace{\Sigma_{ZC}}_{=0} \boldsymbol{\alpha} = \mathbf{0}. \end{aligned} \quad (8)$$

Therefore, $\hat{\boldsymbol{\theta}}_{\text{2SLS}}$ is still biased, but it is consistent to the true value.

2.2.1 2SLS as a Generalized Method of Moments Estimator

In fact, 2SLS is an application of the Generalized Methods of Moments (GMM). For general cases, GMM estimators for parameter $\boldsymbol{\theta}$ (here $\boldsymbol{\theta}$ not only represents the interested parameter in IV regression, but also represents the interested parameter in general cases) are often constructed by

$$\hat{\boldsymbol{\theta}}_{\text{GMM}} = \arg \min_{\boldsymbol{\theta}} \hat{\mathbf{g}}(\boldsymbol{\theta})^\top W \hat{\mathbf{g}}(\boldsymbol{\theta}), \quad (9)$$

where $W \in \mathbb{R}^{m \times m}$ is a positive semi-definite matrix, $\hat{\mathbf{g}}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(X_i, \boldsymbol{\theta}) \in \mathbb{R}^m$ (an estimator of $\mathbf{g}$, and here X_i represents data observation in general cases), $\mathbf{g}(X_i, \boldsymbol{\theta}) \in \mathbb{R}^m$ is a vector-valued function we construct that satisfies

$$\mathbb{E}(\mathbf{g}(X_i, \boldsymbol{\theta}_0)) = \mathbf{0}, \quad (10)$$

where $\boldsymbol{\theta}_0$ is the true value of the parameter (as a variable in (9)) $\boldsymbol{\theta}$. Under mild conditions, we can derive the asymptotic distribution of $\hat{\boldsymbol{\theta}}_{\text{GMM}}$:

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{GMM}} - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, (G^\top W G)^{-1} G^\top W \Omega W^\top G (G^\top W G)^{-1}), \quad (11)$$

where $\Omega = \mathbb{E}(\mathbf{g}(X_i, \boldsymbol{\theta}_0) \mathbf{g}(X_i, \boldsymbol{\theta}_0)^\top)$, $G = \mathbb{E}(\nabla_{\boldsymbol{\theta}} \mathbf{g}(X_i, \boldsymbol{\theta}_0))$. It can be shown that taking $W \propto \Omega^{-1}$ will result in the most efficient estimator of all asymptotically normal estimators with the smallest variance.

For IV regression, we can construct $\mathbf{g}$ as

$$\mathbf{g}_i(\boldsymbol{\theta}) = \mathbf{g}(\mathbf{z}_i, \mathbf{x}_i, y_i, \boldsymbol{\theta}) = \mathbf{z}_i(y_i - \mathbf{x}_i^\top \boldsymbol{\theta}) \in \mathbb{R}^{d_z}. \quad (12)$$

The intuition of this construction is that the estimation of the residual $y_i - \mathbf{x}_i^\top \boldsymbol{\theta}$ is uncorrelated with the instrumental variable $\mathbf{z}_i$, and hence we have

$$\mathbb{E}(\mathbf{g}_i(\boldsymbol{\theta})) = \mathbf{0}. \quad (13)$$

In this case, we have

$$\begin{aligned} \Omega &= \mathbb{E}(\mathbf{g}_i(\boldsymbol{\theta}_0) \mathbf{g}_i(\boldsymbol{\theta}_0)^\top) \\ &= \mathbb{E} \left[\underbrace{(y_i - \mathbf{x}_i^\top \boldsymbol{\theta}_0)^2}_{=\varepsilon_i} \cdot (\mathbf{z}_i \mathbf{z}_i^\top) \right] \\ &= \sigma^2 \Sigma_{ZZ} \end{aligned} \quad (14)$$

and

$$G = \mathbb{E}(\nabla_{\boldsymbol{\theta}} \mathbf{g}_i(\boldsymbol{\theta})) = \mathbb{E}(\mathbf{z}_i \mathbf{x}_i^\top) = \Sigma_{ZX}. \quad (15)$$

Therefore, choosing $W = \Sigma_{ZZ}^{-1} \propto \Omega^{-1}$ leads to the most efficient estimator. However, Σ_{ZZ}^{-1} is unknown, so we replace it with its sample counterpart $S_{ZZ}^{-1} \triangleq (\frac{1}{n} Z^\top Z)^{-1}$, which yields

$$\begin{aligned} \hat{\boldsymbol{\theta}}_{\text{GMM}} &= \arg \min_{\boldsymbol{\theta}} \hat{\mathbf{g}}(\boldsymbol{\theta})^\top W \hat{\mathbf{g}}(\boldsymbol{\theta}) \\ &= \arg \min_{\boldsymbol{\theta}} (S_{ZX} \boldsymbol{\theta} - S_{ZY})^\top S_{ZZ}^{-1} (S_{ZX} \boldsymbol{\theta} - S_{ZY}) \\ &= (S_{XZ} S_{ZZ}^{-1} S_{ZX})^{-1} S_{XZ} S_{ZZ}^{-1} S_{ZY} \\ &= (X^\top P_Z X)^{-1} X^\top P_Z Y = \hat{\boldsymbol{\theta}}_{\text{2SLS}}. \end{aligned} \quad (16)$$

Hence, the 2SLS estimator is equivalent to the GMM estimator with $W = S_{ZZ}^{-1}$. We can also derive its asymptotic distribution via (11), as

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{2\text{SLS}} - \boldsymbol{\theta}_0) \xrightarrow{d} N\left(\mathbf{0}, \sigma^2(\Sigma_{XZ}\Sigma_{ZZ}^{-1}\Sigma_{ZX})^{-1}\right). \quad (17)$$